

HW. Section 3.6

6. Prove there is a prime between n^2 and $(n+1)^2$ for $n=2$ to $n=10$.

n	2	3	4	5	6	7	8	9	10
n^2	4	9	16	25	36	49	64	81	100
p	5	11	17	29	37	53	67	83	101
$(n+1)^2$	9	16	25	36	49	64	81	100	121

24. Use induction to prove the statement $P(n): \sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$ is true for all integers $n \geq 1$

Base Case $n=1$ $P(1)$ is true since $\sum_{i=1}^1 i(i+1) = 1 \cdot 2 = 2 = \frac{1 \cdot 2 \cdot 3}{3} \checkmark$

Induction Step. Let $P(k)$ be true. [Show $P(k+1)$ is true].

$$P(k) \text{ says } \sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$$

$$\text{This means } \sum_{i=1}^{k+1} i(i+1) = \sum_{i=1}^k i(i+1) + (k+1)(k+2) = \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} = \frac{(k+1)(k+2)(k+3)}{3} = \frac{(k+1)((k+1)+1)((k+1)+2)}{3}$$

So $P(k+1)$ is true. whenever $P(k)$ is true

Conclusion Since the statement is true for $n=1$ and is true for $n=k+1$ whenever true for $n=k$ with $k \geq 1$ we conclude by math induction, the statement is true for all integers $n \geq 1$

37. Use induction to prove $\overbrace{n^2 - n}^{P(n)}$ is divisible by 2 for all integers $n \geq 0$

Base case $P(0)$: is true since $0^2 - 0 = 0 = 2 \cdot 0$ is divisible by 2

Induction step. Let $P(k)$ be true [show $P(k+1)$ is true]
 $P(k)$ true says $k^2 - k$ is divisible by 2.

Let $k^2 - k = 2x$ for some $x \in \mathbb{Z}$.

Therefore $(k+1)^2 - (k+1) = k^2 + 2k + 1 - k - 1 = k^2 + k$ is also divisible by 2 since

$$k^2 + k = (2x + k) + k = 2x + 2k = 2(x + k)$$

So $P(k+1)$ is true whenever $P(k)$ is true

Conclusion: Since $P(0)$ is true and since $P(k+1)$ is true whenever $P(k)$ is true for any $k \geq 0$ we conclude by math induction that $P(n)$ is true for all integers $n \geq 0$

68. Use induction to prove $\overbrace{D_x [x^n]}^{P(n)} = n x^{n-1}$ for all integers $n \geq 1$

Base Case $P(1)$ is true since $D_x [x^1] = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{x+h - x}{h} = 1 = 1 \cdot x^{1-1}$

Induction Step Let $P(k)$ be true for $k \geq 1$

This says $D_x [x^k] = k x^{k-1}$

Therefore $D_x [x^{k+1}] = D_x [x^k \cdot x] = x^k \cdot D_x [x] + D_x [x^k] \cdot x$ by product rule

Using Base case that $D_x [x] = 1$ and induction hypothesis that $D_x [x^k] = k x^{k-1}$

$$D_x [x^{k+1}] = x^k \cdot 1 + k x^{k-1} \cdot x = x^k + k x^k = (k+1) x^k$$

So $P(k+1)$ is true whenever $P(k)$ is true.

Conclusion